Fundamentals of Information Systems Security

Chapter 6 Quiz Answer Key

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Section A: Multiple Choice (40 points)

1. **B** — Encryption
2. **B** — Symmetric cryptography
3. **C** — Securely sharing the secret key with the intended recipient
4. **C** — SHA-256
5. **C** — Data integrity verification
6. **C** — Digital signatures
7. **A** — Public Key Infrastructure
8. **B** — Certificate Authority (CA)
9. **C** — AES
10. **B** — To provide authentication, integrity, and nonrepudiation

Section B: True/False (20 points)

1. **True** — Steganography hides message existence
2. **False** — RSA is asymmetric
3. **True** — One-time pad is theoretically unbreakable
4. **False** — MD5 is considered weak and broken
5. **False** — Public keys are shared; private keys are kept secret
6. **True** — TLS uses both asymmetric and symmetric
7. **True** — CA is a trusted entity in PKI
8. **True** — Quantum cryptography uses quantum mechanics
9. **True** — Post-quantum cryptography is quantum-resistant
10. **True** — HMAC uses a secret key with a hash function

Section C: Short Answer (40 points)

1. **Symmetric vs. Asymmetric (10 points)**
   - **Symmetric:** Same key for encryption/decryption. *Strengths:* Very fast, efficient for large data. *Weaknesses:* Key distribution problem, scalability issues. *Examples:* AES, DES, 3DES
   - **Asymmetric:** Public/private key pairs. *Strengths:* Solves key distribution, enables digital signatures. *Weaknesses:* Computationally slow. *Examples:* RSA, ECC
2. **Four Cryptographic Goals (10 points)**
   - **Confidentiality:** Keeping information secret from unauthorized parties
   - **Integrity:** Ensuring information has not been altered
   - **Authentication:** Verifying the identity of an entity
   - **Nonrepudiation:** Preventing an entity from denying an action
3. **Digital Signatures (10 points)**
   - Signer creates a hash of the message
   - Hash is encrypted with the signer's private key
   - Recipient decrypts with the signer's public key
   - Recipient creates a new hash of the received message
   - If hashes match, it proves message came from signer and was not altered
   - **Provides:** Authentication, Integrity, Nonrepudiation
4. **PKI Components (10 points)**
   - **Certificate Authority (CA):** Trusted entity that issues and verifies digital certificates
   - **Registration Authority (RA):** Verifies identity of users requesting certificates
   - **Digital Certificate:** Binds a public key to an identity; contains public key, owner identity, CA signature, validity dates
   - **Certificate Revocation Lists (CRL):** Invalidates certificates before expiry
   - **Online Certificate Status Protocol (OCSP):** Real-time certificate validation